Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)

The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

`daisy`, `dandelion`, `roses`, `sunflowers`, `tulips`.

The images contain variations in:

- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.

Suitable data augmentation may include:

- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.

This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

1 Dataset Description

Table 1: Dataset split by class.

Class	Train	Validation	Test	Total
Daisy	443	95	95	633
Dandelion	629	134	135	898
Roses	449	96	96	641
Sunflowers	489	105	105	699
Tulips	559	120	120	799
Total	2569	550	551	3670

Inference. There are 2,569 training, 550 validation, and 551 test images (3,670 in total). The class counts differ, so macro-averaged metrics are useful in addition to accuracy.

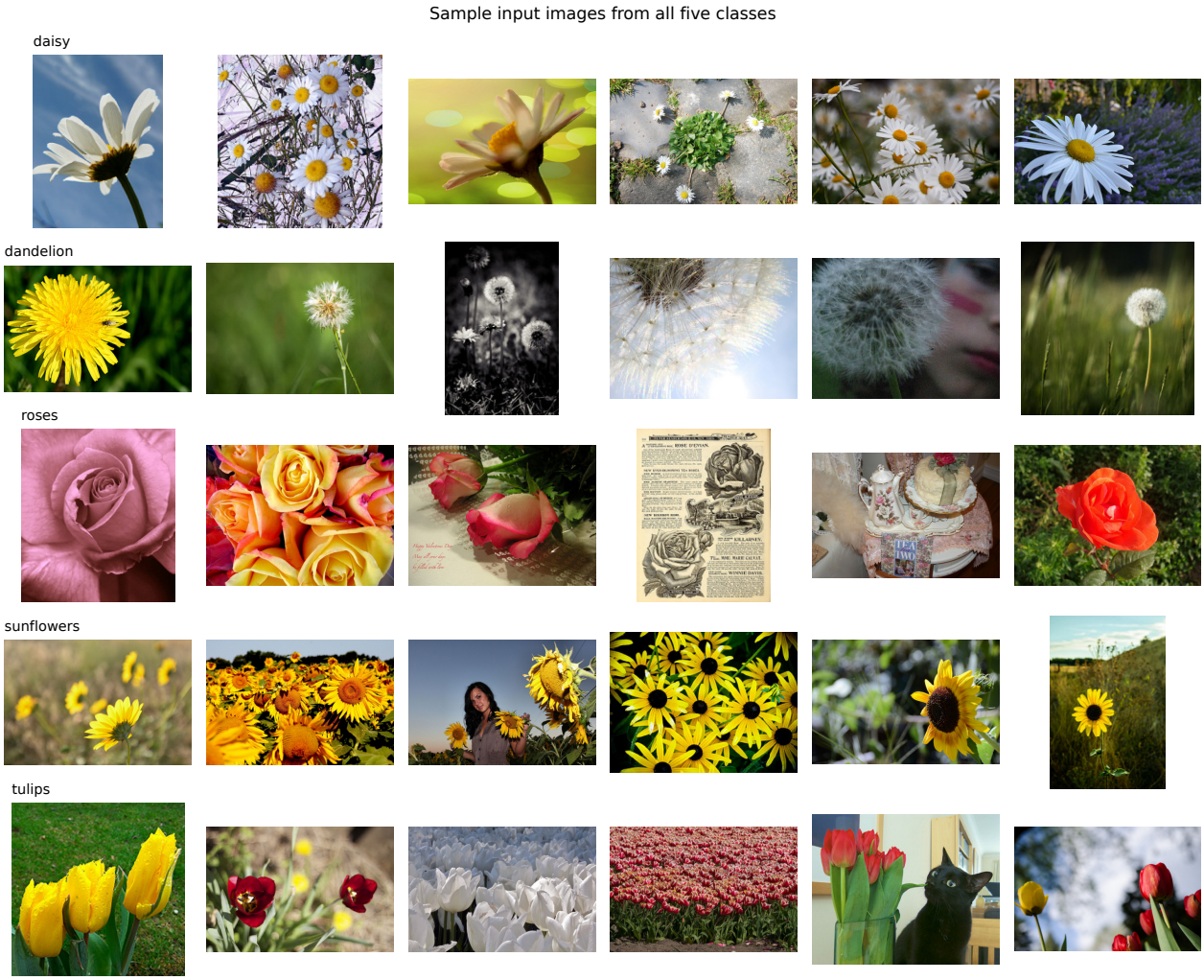


Figure 1: Sample images from all five flower classes.

4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	**Spatial Resolution**	**Channels**
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.
Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.
Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

$$\boxed{\text{Which attention mechanism should be used?}}$$

and

$$\boxed{\text{Where should the attention mechanism be inserted?}}$$

7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1 : \text{ after conv2_x},$$

$$P_2 : \text{ after conv3_x},$$

$$P_3 : \text{ after conv4_x},$$

$$P_4 : \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because P_1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The baseline establishes the reference performance against which the attention-based models are compared.

9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^{C}.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:
- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H \times W \times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\max} = \mathrm{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\mathrm{avg}} \| F_{\max}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\mathrm{Conv}_{7\times 7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H\times W\times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \to \text{Channel Attention} \to \text{Spatial Attention} \to F_{cs}}$$

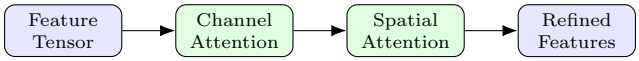
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H\times W\times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N \times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\text{SA}} = F + \gamma\, \text{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

$$\boxed{\text{Which spatial features are related to other spatial features?}}$$

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

13. Inserting Attention into ResNet50

For any selected stage P_k, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

The student must record the feature dimensions immediately before and after the attention block.
Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.

14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.
Compare the following:

Model	**Channel**	**Spatial**	**Self-Attention**
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates:

$$\boxed{\text{How does attention type affect model performance?}}$$

Students must keep the following identical:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- learning rate schedule;
- epochs;
- batch size;
- backbone;
- classifier;
- attention position; and
- evaluation procedure.

2 Main Quantitative Results

Table 2: Recorded test results.

Run	Attention	Accuracy	Macro P	Macro R	Macro F1	Time (s)
B1_ChannelSpatial_P1	Channel+Spatial	0.961887	0.960974	0.962088	0.961494	938.3190
A3_ChannelSpatial_P3	Channel+Spatial	0.960073	0.959907	0.960161	0.959994	770.1499
B2_ChannelSpatial_P2	Channel+Spatial	0.958258	0.960176	0.957536	0.958618	902.2237
A4_SelfAttn_P3	Self-Attention	0.954628	0.954709	0.953704	0.954120	1006.009
A0_Baseline	Baseline	0.952813	0.954703	0.951308	0.952599	976.6588
A1_Channel_P3	Channel (SE)	0.952813	0.952543	0.952247	0.952038	866.3085
A2_Spatial_P3	Spatial	0.947368	0.949875	0.945082	0.947149	1018.936
A5_Combined_P3	Combined	0.943739	0.947726	0.939972	0.942798	986.2221
E1_Kernel3_P3	Channel+Spatial*	0.920145	0.922855	0.916960	0.918957	251.3754
F1_ECA_P3	ECA	0.912886	0.916514	0.909522	0.912198	335.3402
E2_Kernel5_P3	Channel+Spatial*	0.909256	0.909962	0.907734	0.908673	274.9477
F3_CoordAtt_P3	Coordinate Attention	0.909256	0.909186	0.909117	0.908828	323.0334
B4_ChannelSpatial_P4	Channel+Spatial	0.907441	0.910636	0.902841	0.905904	279.9083
C2_ChannelSpatial_P2P3P4	Channel+Spatial	0.905626	0.912421	0.901059	0.904020	316.7446
C1_ChannelSpatial_P3P4	Channel+Spatial	0.903811	0.907279	0.900846	0.903303	312.7413
D1_SpatialChannel_P3	Spatial→Channel	0.891107	0.902599	0.885018	0.889186	282.7467
F2_CBAM_P3	CBAM	0.891107	0.896379	0.886817	0.889019	323.5703

Inference. B1/P1 has the highest listed accuracy (0.961887) and macro F1 (0.961494). A3/P3 has accuracy 0.960073 and macro F1 0.959994, compared with baseline values of 0.952813 and 0.952599. In these recorded runs, P4 and multi-position results are lower than P3. ECA is higher than CBAM and Coordinate Attention on both accuracy and macro F1.

2.1 Accuracy and Macro-F1 Graphs

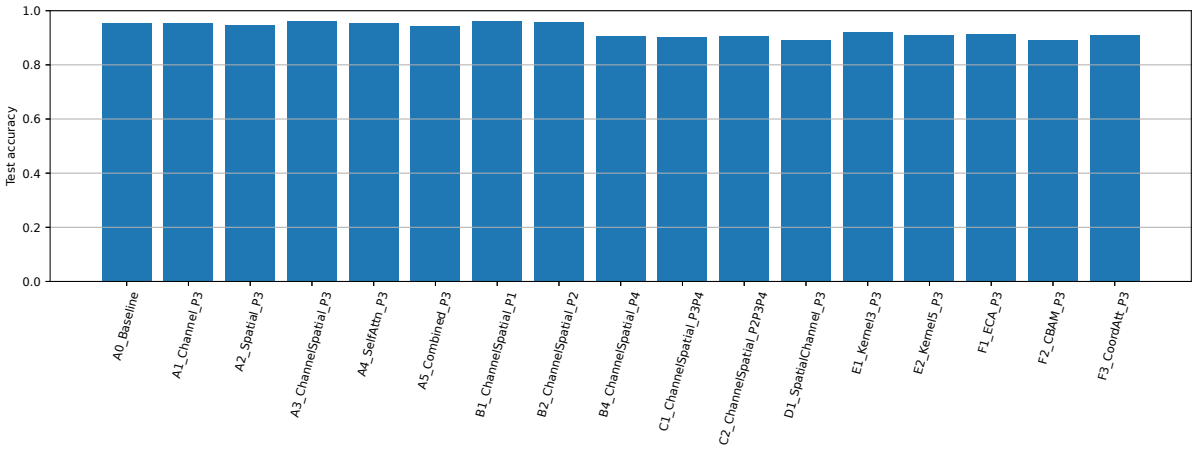


Figure 2: Test accuracy across recorded runs.

Inference. The table indicates B1 has the highest accuracy, followed by A3 and B2.

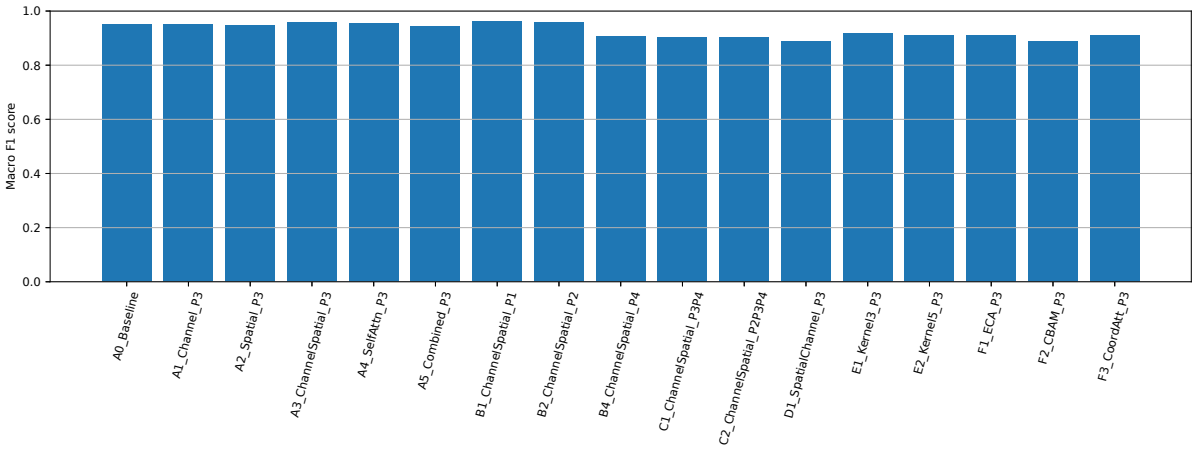


Figure 3: Macro F1 across recorded runs.

Inference. B1 has the highest listed macro F1, followed by A3 and B2. Macro F1 complements accuracy by averaging class-level F1 scores.

3 Training and Validation Curves

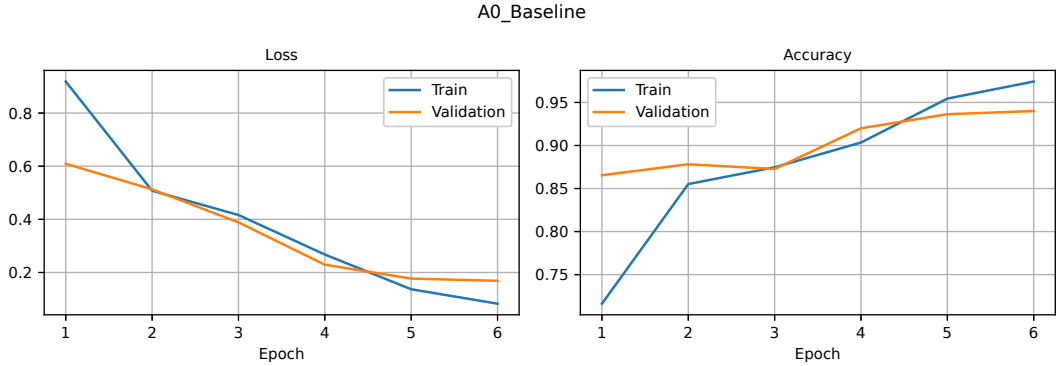


Figure 4: Baseline training/validation curve.

Inference. Training and validation accuracy generally increase over the epochs, while both loss curves decrease. The validation curves follow the training trends reasonably closely, suggesting learning progress without an obvious large gap in this run.

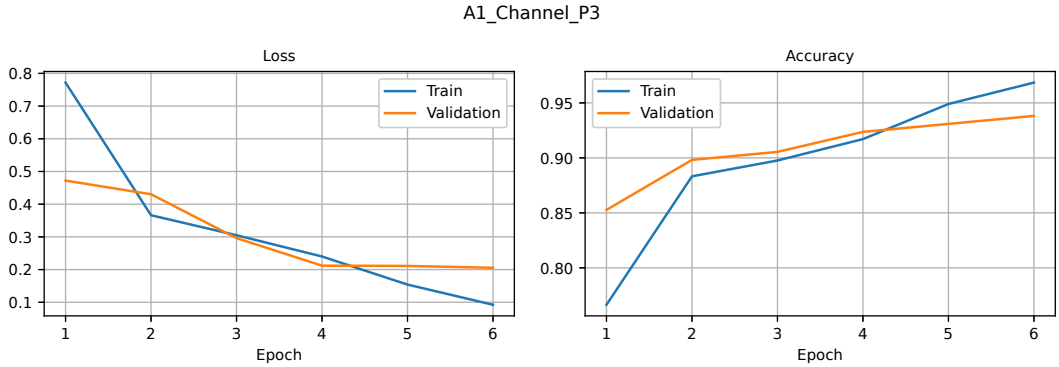


Figure 5: Channel attention at P3.

Inference. Accuracy increases and loss decreases across training, indicating that the model learns useful flower features. The validation curves broadly follow the training curves, although the plot alone does not establish that channel attention is better than the baseline.

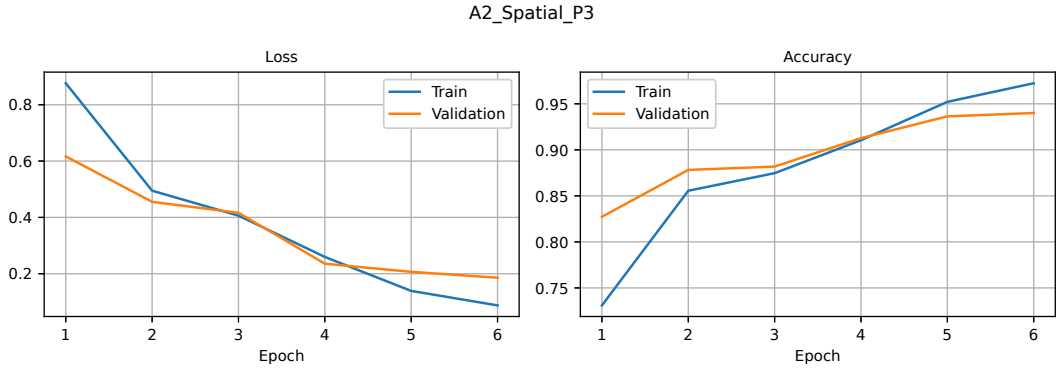


Figure 6: Spatial attention at P3.

Inference. Training and validation accuracy rise while their losses decline, showing that the model learns during the experiment. Its final test performance is lower than the baseline in the recorded results, so improving convergence does not necessarily mean improving classification performance.

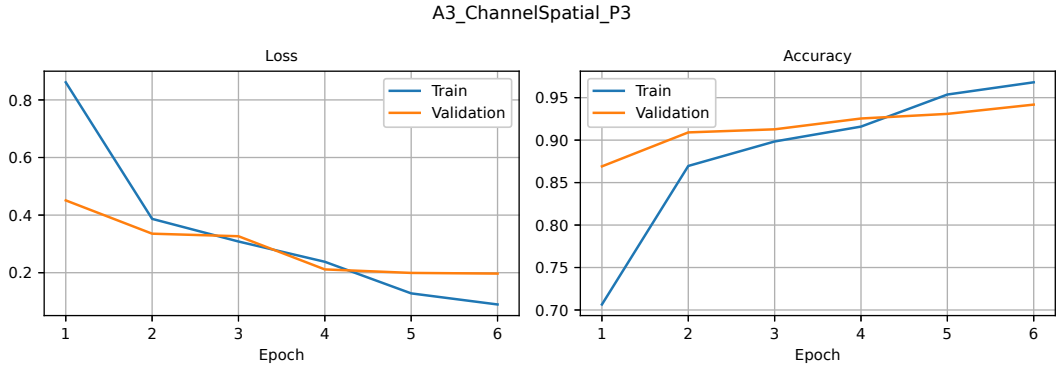


Figure 7: Channel–spatial attention at P3.

Inference. The accuracy curves increase and the loss curves decrease over the displayed epochs. The recorded test accuracy is 0.960073, higher than the baseline's 0.952813, suggesting a performance improvement in these runs.

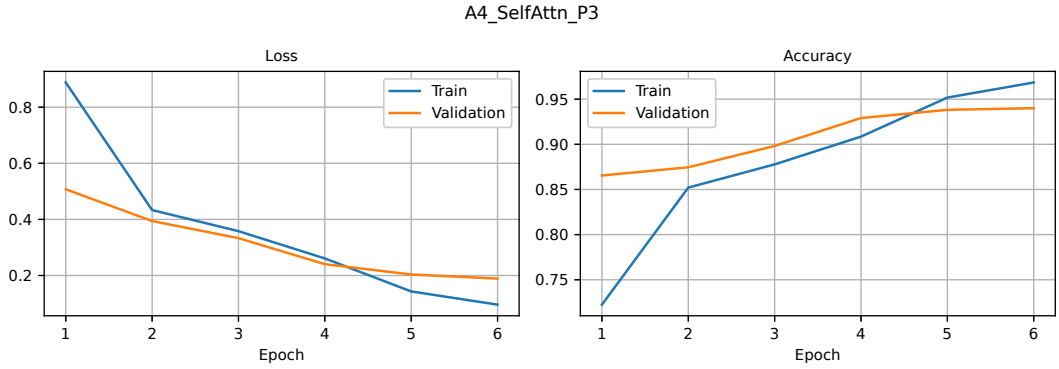


Figure 8: Self-attention at P3.

Inference. The plotted accuracy increases as loss decreases, indicating learning progress. Its recorded test accuracy is 0.954628, slightly above the baseline, but the curves alone cannot show whether self-attention caused that improvement.

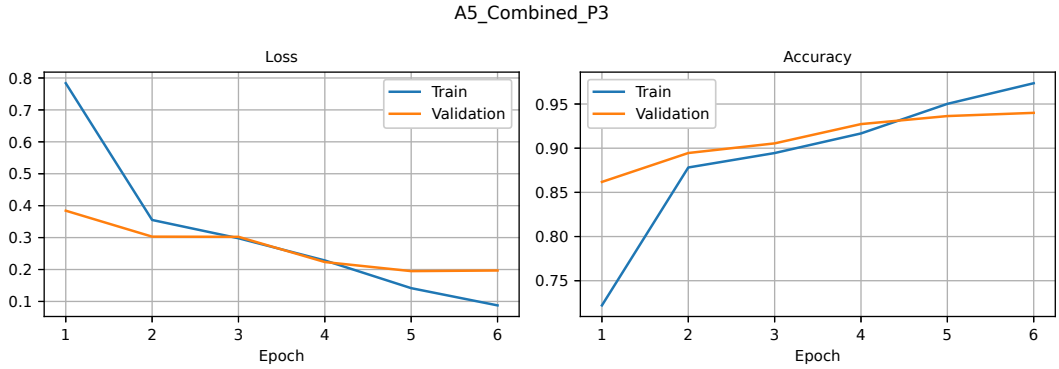


Figure 9: Combined attention at P3.

Inference. Training and validation accuracy improve, and the losses decline throughout the displayed epochs. However, its test accuracy (0.943739) is below the baseline, suggesting that combining attention mechanisms does not automatically improve classification.

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

$$\boxed{\text{Channel Attention} \rightarrow \text{Spatial Attention}}$$

and insert exactly the same module at different stages.

Experiment	Attention Position	Resolution	Channels
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

Students should study whether attention is more beneficial for:

- high-resolution low-level features;
- intermediate representations; or
- low-resolution semantic features.

The principal question is:

$$\boxed{\text{How does the location of attention influence performance?}}$$

4 Attention-Position Ablation

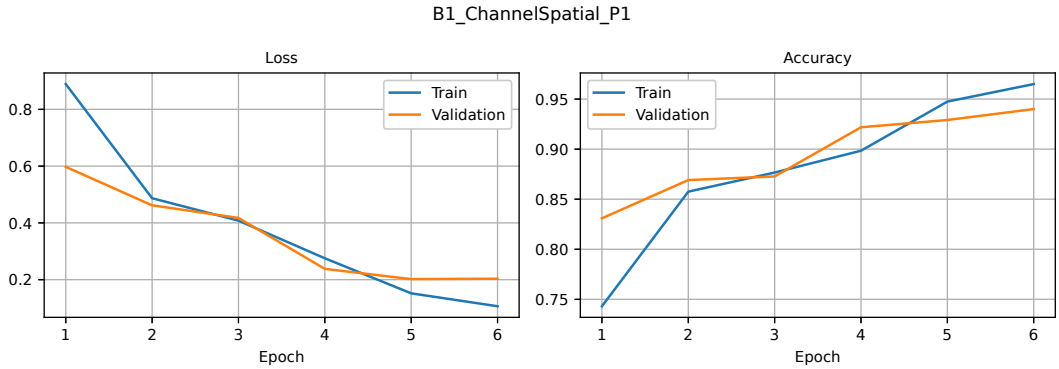


Figure 10: Channel–spatial attention at P1.

Inference. The curves show increasing accuracy and decreasing loss during the displayed training. The recorded test accuracy is 0.961887, the highest among the listed P1–P4 position runs, although the P1 result should be interpreted alongside its greater feature-map resolution and computational cost.

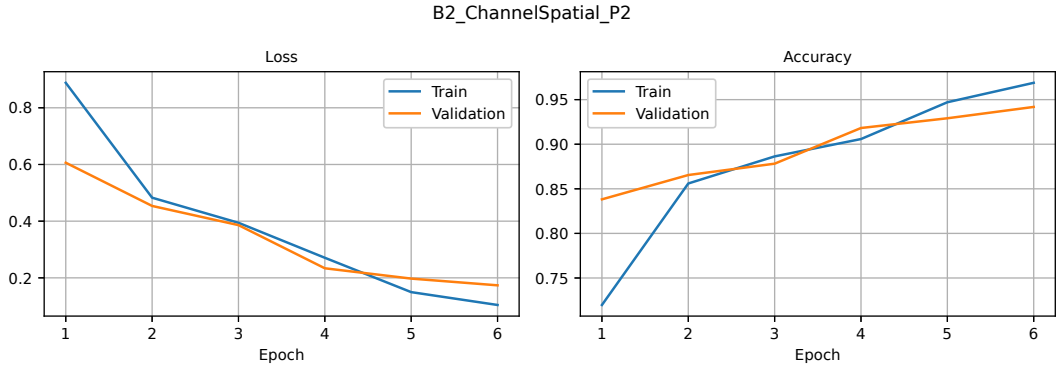


Figure 11: Channel–spatial attention at P2.

Inference. Training progresses with increasing accuracy and decreasing loss. Its test accuracy is 0.958258, close to P3's 0.960073, indicating that both intermediate positions perform well in the recorded results.

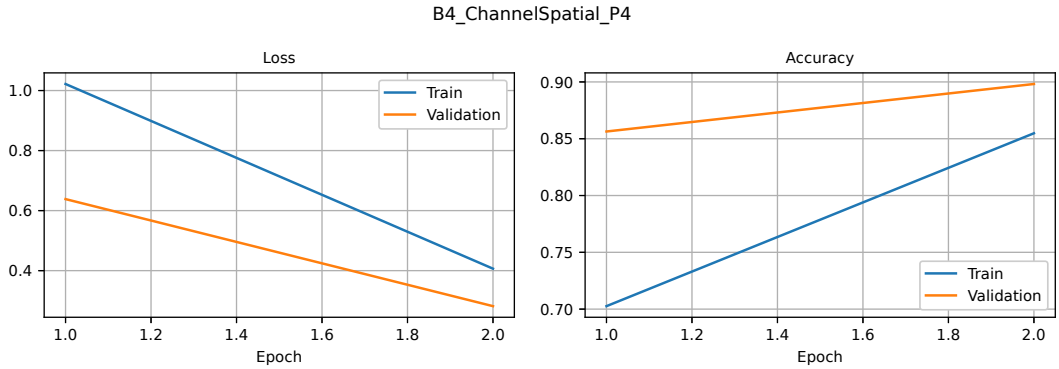


Figure 12: Channel–spatial attention at P4.

Inference. The short curves show improving training accuracy and declining loss, but the displayed run has very few epochs. Its recorded test accuracy (0.907441) is lower than P1, P2, and P3; confirm the training protocol before attributing this difference to position alone.

16. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage.
Compare:

$$\text{Attention only at } P_3,$$

$$\text{Attention only at } P_4,$$

and

$$\text{Attention at } P_3 + P_4.$$

As an extended experiment, students may evaluate:

$$P_2 + P_3 + P_4.$$

The purpose is to investigate whether applying attention repeatedly provides sufficient performance improvement to justify the additional:

- parameters;

- computation;
- memory;
- training time; and
- inference time.

Students should experimentally answer:

Is attention at more locations necessarily more effective?

5 Multiple Attention Positions

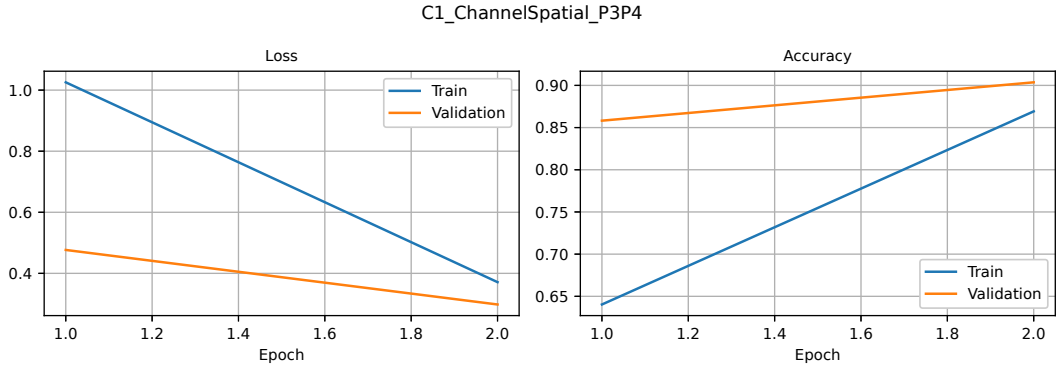


Figure 13: Attention at P3 and P4.

Inference. The plotted training accuracy increases and loss decreases over the short run. Its test accuracy is 0.903811, substantially below the single-P3 result, so adding attention at a second position did not improve performance in these recorded experiments.

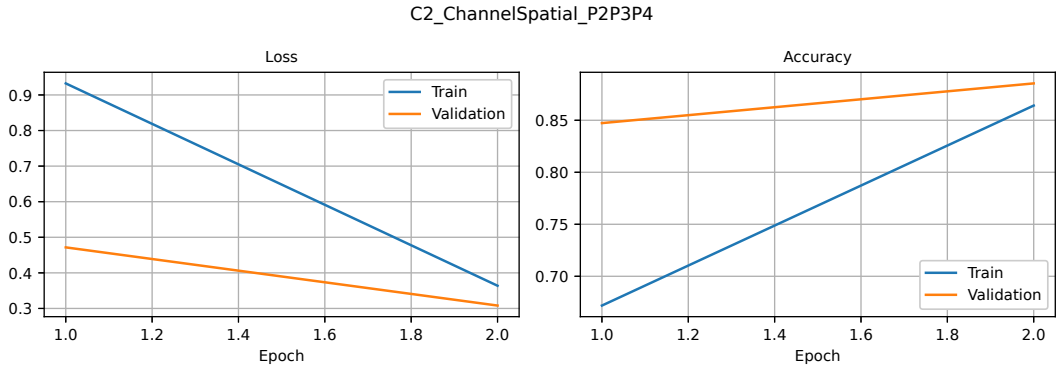


Figure 14: Attention at P2, P3, and P4.

Inference. Accuracy rises and loss declines during the displayed training, but the run is short. Its test accuracy (0.905626) is slightly higher than P3+P4, yet both multi-position results are below single-P3 performance.

17. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features.
Compare:

Channel → Spatial

against

Spatial → Channel

while maintaining the same insertion position.

For example, at P_3:

$$F \to M_c \to M_s$$

should be compared with

$$F \to M_s \to M_c.$$

6 Attention Ordering

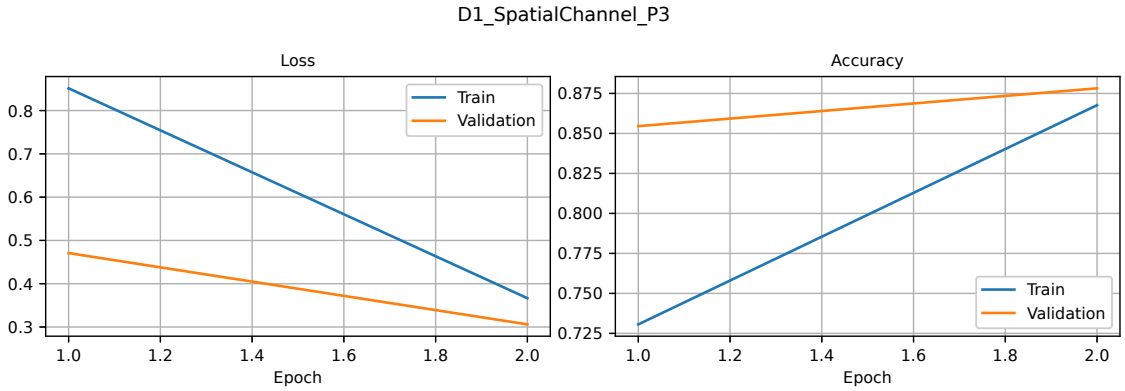


Figure 15: Spatial then channel attention at P3.

Inference. The curves show training progress, with accuracy increasing and loss decreasing. Its recorded test accuracy is 0.891107, compared with 0.960073 for Channel → Spatial at P3; this favours Channel → Spatial if the other experimental settings were identical.

18. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result.
Evaluate:

$$k \in \{3, 5, 7\}.$$

Table 3: Available kernel-size results.

Kernel	Accuracy	Macro F1	Time (s)
3×3	0.920145	0.918957	251.3754
5×5	0.909256	0.908673	274.9477
7×7	Not provided	Not provided	Not provided

Inference. Among the two provided rows, kernel 3 has higher accuracy and macro F1 than kernel 5. The required 7×7 result is missing. Verify that the kernel was the only change before treating this as a kernel-size ablation.

7 Spatial-Attention Kernel Size

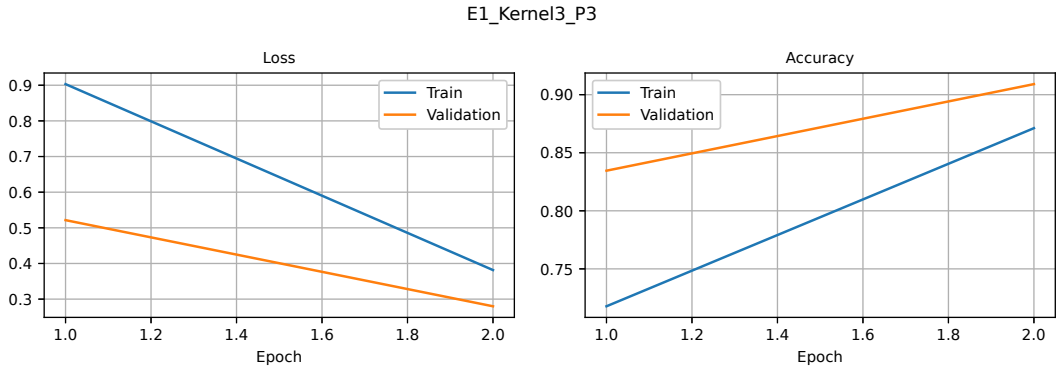


Figure 16: Kernel 3 run at P3.

Inference. The training curves show accuracy increasing and loss decreasing over the short run. The recorded test accuracy is 0.920145, higher than the 5×5 result, but the missing 7×7 result prevents a complete kernel-size comparison.

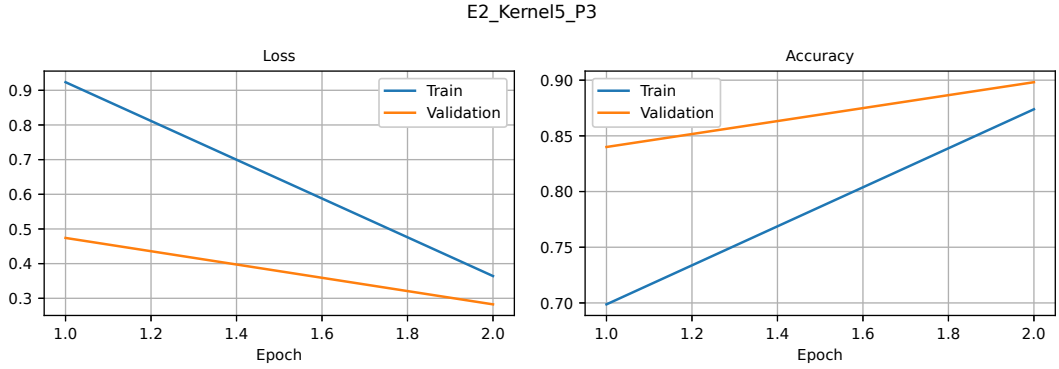


Figure 17: Kernel 5 run at P3.

Inference. Accuracy increases and loss decreases during the displayed training. Its recorded test accuracy (0.909256) and macro F1 (0.908673) are lower than the 3×3 results; do not conclude that smaller kernels are always better from just these two runs.

19. Additional Task: Comparison of Different Attention Mechanisms

As an additional exploration task, students should compare the attention mechanisms implemented in the laboratory with established attention approaches.

Students should select at least

two additional attention mechanisms

from the following:

- Squeeze-and-Excitation (SE) Attention;
- Efficient Channel Attention (ECA);
- Convolutional Block Attention Module (CBAM);
- Coordinate Attention; and
- lightweight Self-Attention.

The pretrained backbone must remain:

ResNet50

for all comparisons.

To ensure a fair comparison, additional attention mechanisms should preferably be inserted at the same position:

$P_3 = \text{after conv4_x}$

8 Additional Attention Mechanisms

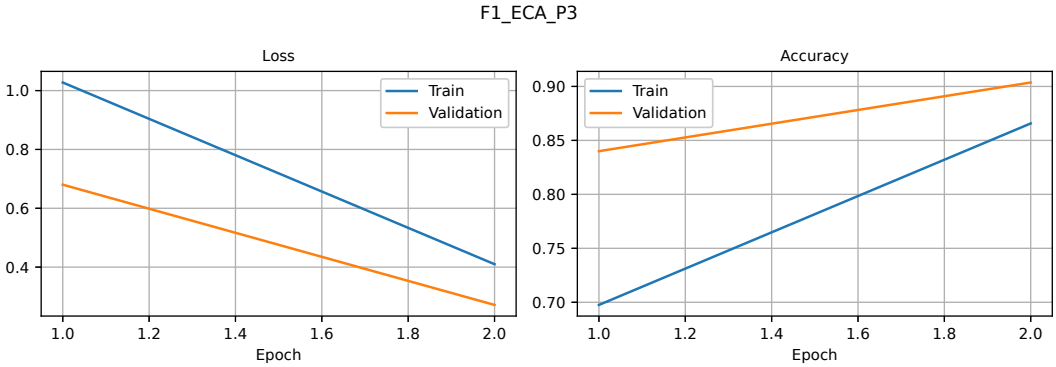


Figure 18: ECA at P3.

Inference. The curves indicate improving training accuracy and decreasing loss during the displayed run. ECA records test accuracy 0.912886 and macro F1 0.912198; its parameter count is missing, so its efficiency cannot be fully compared with the other mechanisms.

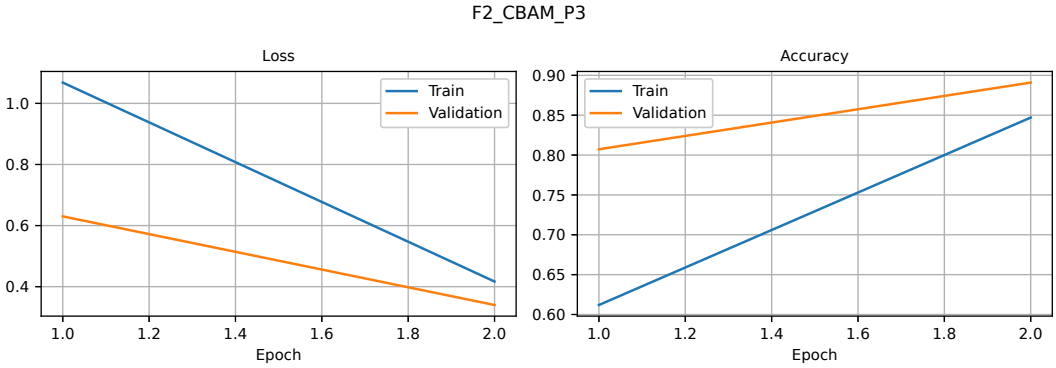


Figure 19: CBAM at P3.

Inference. The curves show learning progress, with accuracy increasing and loss decreasing. Its test accuracy (0.891107) is lower than the recorded ECA and Coordinate Attention results, but parameter counts are needed to compare model complexity fairly.

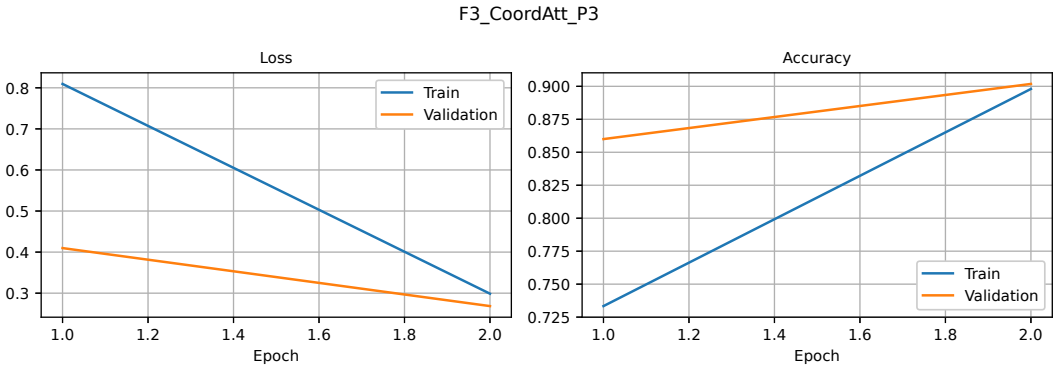


Figure 20: Coordinate Attention at P3.

Inference. Training accuracy rises and loss decreases during the displayed run. Its test accuracy is 0.909256 and macro F1 is 0.908828, both below ECA's recorded values but above CBAM's.

19.1 Suggested Comparison

Table 4: Comparison of attention mechanisms using the recorded test results.

Attention	Accuracy	Precision	Recall	Macro F1	Inference Time (s)
No Attention / Baseline	0.952813	0.954703	0.951308	0.952599	976.6588
Channel / SE	0.952813	0.952543	0.952247	0.952038	866.3085
Spatial Attention	0.947368	0.949875	0.945082	0.947149	1018.9360
ECA	0.912886	0.916514	0.909522	0.912198	335.3402
CBAM	0.891107	0.896379	0.886817	0.889019	323.5703
Self-Attention	0.954628	0.954709	0.953704	0.954120	1006.0090
Coordinate Attention	0.909256	0.909186	0.909117	0.908828	323.0334

Inference. Among the listed mechanisms, Self-Attention has the highest recorded accuracy (0.954628) and macro F1 (0.954120). These are descriptive results; parameter counts are unavailable, and the timing unit needs confirmation.

19.2 Questions for Attention-Mechanism Comparison

Students should investigate:

1. How does classification accuracy vary across attention mechanisms?
2. How does macro F1-score vary across attention mechanisms?
3. How many parameters are introduced by each attention module?
4. How does each attention mechanism influence inference time?
5. Does attention improve the convergence behavior?
6. Does attention produce more focused feature localization?
7. Does an increase in attention complexity always improve classification performance?
8. Does a lightweight attention mechanism provide comparable performance with a computationally heavier mechanism?

The objective is to understand:

$$\boxed{\text{Performance} + \text{Complexity} + \text{Attention Behaviour}}$$

rather than considering only classification accuracy.

20. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.

The exact values selected by the student should remain identical for all controlled comparisons.

21. Loss Function

For the five-class classification problem, use categorical cross-entropy.
For N samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.
The same loss formulation must be used for every attention comparison.

22. Evaluation Metrics

Students must report the following quantitative metrics:

- classification accuracy;
- macro precision;
- macro recall;
- macro F1-score;
- number of trainable parameters;
- total number of parameters; and
- average inference time per image.

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\frac{\text{Precision}_c\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

23. Required Main Result Table

Inference. Among the recorded channel–spatial position runs, P_3 has the highest accuracy (0.960073), followed by P_2 (0.958258), while P_4 is lower (0.907441). Confirm that only insertion position changed and that each model follows the required architecture before making a causal claim.

24. Graphical Performance Analysis

Students should not present only numerical tables.
Appropriate graphs must be generated to analyze model behavior.
At least three of the following graphical comparisons should be included in the laboratory report.

Table 5: Comparison of models and attention insertion positions.

Model	Position	Accuracy	Precision	Recall	F1	Parameters	Inference Time (s)
Baseline	–	0.952813	0.954703	0.951308	0.952599	N/A	976.6588
Channel Attention	P_3	0.952813	0.952543	0.952247	0.952038	N/A	866.3085
Spatial Attention	P_3	0.947368	0.949875	0.945082	0.947149	N/A	1018.9360
Channel + Spatial	P_2	0.958258	0.960176	0.957536	0.958618	N/A	902.2237
Channel + Spatial	P_3	0.960073	0.959907	0.960161	0.959994	N/A	770.1499
Channel + Spatial	P_4	0.907441	0.910636	0.902841	0.905904	N/A	279.9083
Self-Attention	P_3	0.954628	0.954709	0.953704	0.954120	N/A	1006.0090
ECA (Additional Attention 1)	P_3	0.912886	0.916514	0.909522	0.912198	N/A	335.3402
CBAM (Additional Attention 2)	P_3	0.891107	0.896379	0.886817	0.889019	N/A	323.5703

24.1 Training and Validation Accuracy

Plot

$$\text{Epoch} \quad \text{versus} \quad \text{Accuracy}$$

for:

- baseline ResNet50; and
- selected attention models.

Both training and validation accuracy should be shown.

Students should comment on whether attention:

- accelerates convergence;
- produces smoother learning;
- reduces the train–validation gap;
- leads to overfitting; or
- causes unstable training.

24.2 Training and Validation Loss

Plot

$$\text{Epoch} \quad \text{versus} \quad \mathcal{L}_{CE}.$$

The graph should include training and validation loss.

Students should compare the convergence behavior of the baseline and attention models.

24.3 Attention Mechanism versus Accuracy

Generate a bar graph:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Classification Accuracy}.$$

Possible entries are:

$$\text{Baseline, Channel, Spatial, Channel+Spatial, Self-Attention, SE, ECA, CBAM}.$$

24.4 Attention Mechanism versus Macro F1

Similarly, plot:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Macro F1-score.}$$

This comparison is useful when the class-wise performance is not uniform.

24.5 Accuracy versus Model Parameters

Create a scatter plot using:

$$x = \text{Total Model Parameters}$$

and

$$y = \text{Test Accuracy.}$$

Each point should represent one attention mechanism.
This allows the student to study the

$$\boxed{\text{Accuracy–Complexity Trade-off}}$$

instead of evaluating accuracy independently.

24.6 Macro F1 versus Inference Time

Plot:

$$x = \text{Inference Time per Image}$$

and

$$y = \text{Macro F1-score.}$$

Students should analyze whether additional computational cost produces a meaningful increase in classification performance.

25. Confusion Matrix Analysis

Generate a confusion matrix for:
1. baseline ResNet50; and
2. the selected attention-based ResNet50.

9 Confusion Matrix and Class-Wise Results

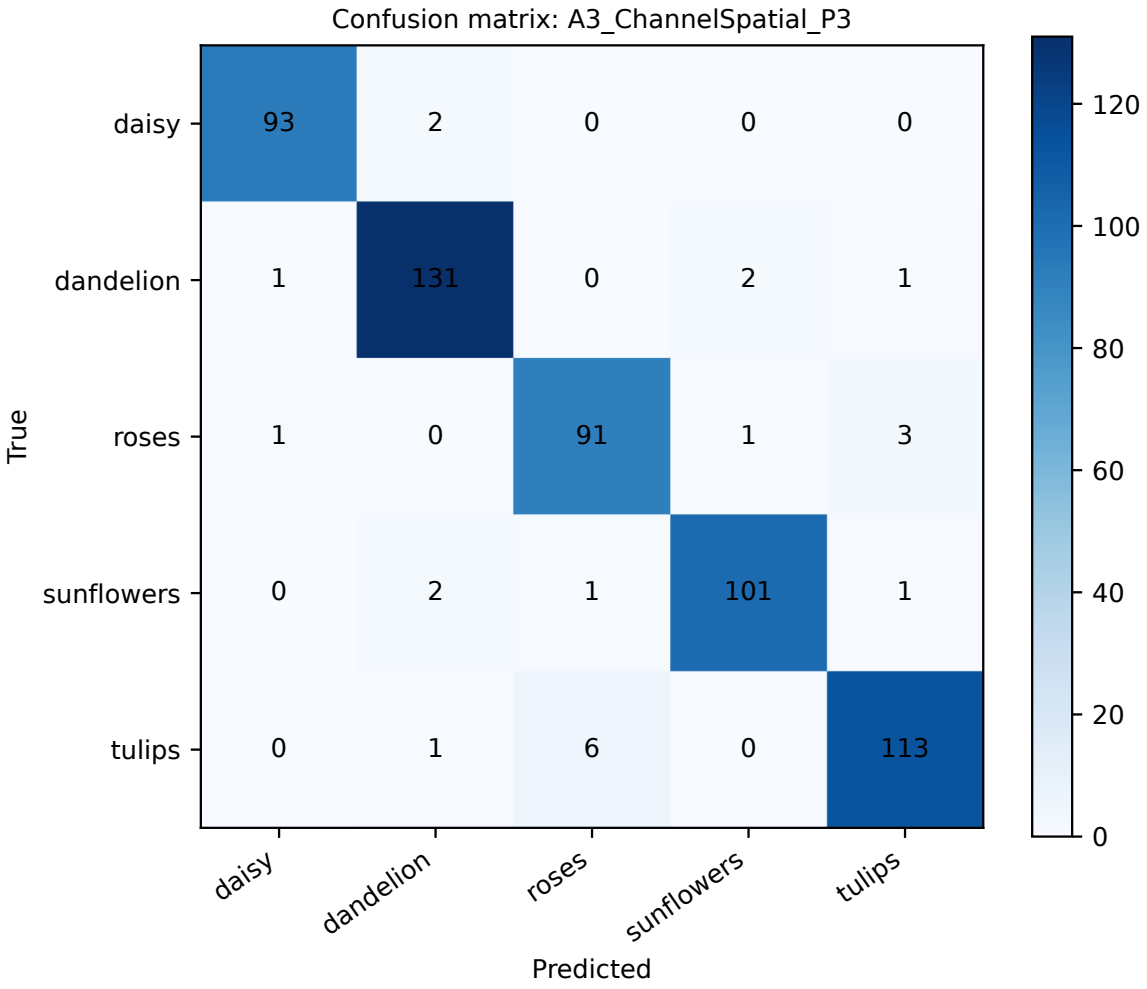


Figure 21: Confusion matrix for channel–spatial attention at P3.

Inference. The model correctly classifies most test images, with 93 daisies, 131 dandelions, 91 roses, 101 sunflowers, and 113 tulips classified correctly. The most visible confusion is between tulips and roses, with six tulips predicted as roses.

Table 6: Class-wise test report for A3/P3.

Class	Precision	Recall	F1-score	Support
Daisy	0.978947	0.978947	0.978947	95
Dandelion	0.963235	0.970370	0.966679	135
Roses	0.928571	0.947917	0.938144	96
Sunflowers	0.971154	0.961905	0.966507	105
Tulips	0.957627	0.941667	0.949580	120
Macro average	0.959907	0.960161	0.959994	551
Weighted average	0.960192	0.960073	0.960093	551

Inference. Daisy has the highest class F1 (0.978947), while roses has the lowest (0.938144) in this report. Tulips has the lowest recall (0.941667). Inspect the confusion matrix itself to name the exact pairs of classes that are confused; do not infer those pairs from the classification report alone.

26. Qualitative Analysis of Attention

Quantitative classification performance does not directly explain where the network obtains evidence for its prediction.

Students must therefore perform qualitative analysis.

The visual analysis should include:

- internal attention maps;
- attention overlays;
- intermediate feature maps; and
- Grad-CAM explanations.

Students should preferably use the same input image when comparing different models.

27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction.

Let

$$A^k$$

represent feature map k from a selected convolutional layer.

For class score y^c, the importance coefficient is:

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is:

$$L_{\text{Grad}-\text{CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

The map should be resized to the original image resolution and overlaid on the input.

Students should compare:

Input | Baseline Grad-CAM | Attention Model Grad-CAM

for the same image.

10 Grad-CAM and Attention Activation



Figure 22: Grad-CAM for channel–spatial attention at P3.

27.1 Grad-CAM Interpretation

Students should qualitatively determine whether the attention model:

- focuses more strongly on the flower;
- suppresses irrelevant background regions;
- concentrates on petals;
- concentrates on the central flower structure;
- attends to discriminative edges;
- distributes activation across the entire flower;
- incorrectly focuses on leaves;
- incorrectly focuses on background objects; or
- changes its focus for an incorrect prediction.

Students should avoid unsupported statements such as:

> "The Grad-CAM of Model A is better."

Instead, the observation should describe visible evidence.
For example:

> The baseline network activates both the flower and surrounding leaves, whereas the attention-based model produces stronger activation over the petal region and weaker activation over the background.

28. Internal Attention Maps

An important objective of this experiment is to distinguish between:

Internal Attention

and

Post-hoc Explanation

because they are not the same.
An attention module directly computes weights during the forward pass.
Grad-CAM, in contrast, is generated after prediction using gradients.

28.1 Channel-Attention Visualization

For channel attention:

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

Students should visualize the learned channel importance values using a bar graph.
Instead of displaying every channel when C is large, the student may show the top-K channels.
Recommended values are:

$$K = 10$$

or

$$K = 20.$$

Students should comment on whether:
- a small number of channels receive strong weights;
- many channels contribute similarly; or
- channel weighting changes substantially for different input images.

28.2 Spatial-Attention Visualization

For spatial attention:

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

Resize the attention map to:

$$224 \times 224$$

and overlay it on the input image.
Display:

$$\boxed{\text{Input} \mid \text{Spatial Attention Map} \mid \text{Attention Overlay}}$$

Students should identify the image regions receiving high attention values.

28.3 Self-Attention Visualization

For self-attention, students may visualize one of the following:
- attention associated with a selected query location;
- mean attention across query positions;
- maximum attention across query positions; or
- an aggregated spatial self-attention response.

The visualization should be reshaped into a spatial form and, when appropriate, resized to the image resolution.
Students should analyze how distant regions interact within the feature map.

29. Attention Position Visualization

Use the same test image to visualize spatial attention at different ResNet50 stages.
Generate:

$$M_s^{P_2}, \qquad M_s^{P_3}, \qquad M_s^{P_4}.$$

Display the maps as:

$$\boxed{\text{Input} \mid M_s^{P_2} \mid M_s^{P_3} \mid M_s^{P_4}}$$

The objective is to visually study how attention changes as feature representations become deeper.
Possible observations may include:

- P_2 responding strongly to edges, colors, textures, and fine structures;
- P_3 responding to flower parts and intermediate object structures;
- P_4 responding to larger semantic flower regions.

These observations must be based on the student's experimental visualizations.

30. Qualitative Comparison Across Attention Mechanisms

Select at least:

3 correctly classified images

and

2 incorrectly classified images.

For every selected image, compare the behavior of multiple models.
A recommended arrangement is:

Input Image
↓
Baseline Grad-CAM
Channel-Attention Grad-CAM
Spatial-Attention Map
Channel+Spatial Grad-CAM
Self-Attention Response

Students may alternatively arrange the images horizontally in the report.

30.1 Qualitative Criteria

The student should discuss:

- foreground localization;
- background suppression;
- localization of petals;
- localization of the flower center;
- activation compactness;
- activation coverage;
- incorrectly attended regions;
- differences between correctly and incorrectly classified samples;
- differences between shallow and deep attention; and
- consistency of attention across different images.

31. Required Visualizations

The laboratory report must contain the following visualizations:

1. sample input images from all five classes;
2. training and validation accuracy curves;
3. training and validation loss curves;
4. confusion matrix;
5. feature map before attention;
6. channel-attention weights;
7. spatial-attention heatmap;
8. spatial-attention overlay;
9. self-attention response for the self-attention experiment;
10. feature map after attention;
11. baseline Grad-CAM;
12. Grad-CAM for at least one attention-based model;
13. attention maps at P_2, P_3, and P_4; and
14. at least one graphical comparison of the performances of different attention mechanisms.

32. Recommended Visualization Layout

For one representative test image, students are encouraged to generate a figure in the following format:

Input	Baseline Grad-CAM	Channel Grad-CAM	Spatial Attention Map	Channel+Spatial Grad-CAM

For students completing the additional attention comparison, the figure may be extended to:

Input|Baseline|SE|ECA|CBAM|Self-Attention

using appropriate visualization methods.

33. Required Experimental Analysis

The final report should contain three forms of analysis.

33.1 Quantitative Analysis

Report:

Accuracy + Precision + Recall + Macro F1

33.2 Complexity Analysis

Report:

Parameters + Inference Time

33.3 Qualitative Analysis

Report:

Grad-CAM + Attention Maps + Visual Interpretation

Therefore, the overall evaluation should be:

Quantitative + Computational + Qualitative

34. Experimental Protocol

To ensure fair comparison, all ablation experiments must use:

- identical training images;
- identical validation images;
- identical testing images;
- identical random seed where possible;
- identical image preprocessing;
- identical augmentation;
- identical backbone;
- identical classifier;
- identical optimizer;
- identical batch size;
- identical number of epochs;
- identical learning-rate schedule;
- identical fine-tuning procedure; and
- identical evaluation metrics.

Only the experimental factor under investigation should change.
For example, when studying attention position:

Only Attention Position Should Change

and when studying attention type:

Only Attention Type Should Change.

35. Minimum Mandatory Ablation Set

If laboratory time is limited, every student must complete at least the following experiments:

1. Baseline pretrained ResNet50.
2. Channel attention at P_3.
3. Spatial attention at P_3.
4. Channel+Spatial attention at P_3.
5. Channel+Spatial attention at P_2.
6. Channel+Spatial attention at P_4.
7. Self-attention at P_3.
8. Grad-CAM comparison between baseline ResNet50 and one selected attention model.
9. Spatial-attention visualization at P_2, P_3, and P_4.
10. Graphical comparison of the classification performance of the attention models.

This minimum set allows students to study:

Attention Type

and

Attention Position.

36. Additional Student Exploration

Students interested in extended experimentation should select at least two mechanisms from:

SE, ECA, CBAM, Coordinate Attention, Self-Attention

and integrate them with the same ResNet50 backbone.
Students should maintain the same:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- epochs;
- fine-tuning strategy; and
- attention position.

The additional investigation should include both:

Quantitative Comparison

and

Qualitative Comparison.

37. Questions Students Must Answer

1. **What is the purpose of residual connections in ResNet50?**
   Residual connections let information and gradients pass through shortcut paths. This helps train a deep network and reduces the vanishing-gradient problem.

2. **What does `conv2_x` represent compared with `conv5_x`?**
   `conv2_x` produces earlier, higher-resolution features such as edges, colours, and textures. `conv5_x` produces smaller, lower-resolution feature maps with more high-level semantic information.
3. **Why does the spatial resolution decrease with network depth?**
   Downsampling operations reduce the height and width of feature maps as the network gets deeper. This reduces computation and helps the network represent larger patterns.
4. **Why does the number of feature channels increase with network depth?**
   Deeper layers combine earlier features into more complex patterns. More channels give the network capacity to represent a wider range of these patterns.
5. **What is meant by low-level, intermediate, and high-level features?**
   Low-level features include edges, colours, and simple textures. Intermediate features represent parts and shapes, while high-level features represent meaningful object or class information.
6. **What is the purpose of an attention mechanism?**
   Attention assigns different importance to features or image locations. It helps the model emphasize information useful for making a prediction.
7. **What does channel attention attempt to learn?**
   Channel attention learns which feature channels are more useful. It assigns weights to channels and strengthens or reduces their contribution.
8. **What does spatial attention attempt to learn?**
   Spatial attention learns which locations in a feature map are important. It creates a spatial weight map to emphasize useful regions and reduce less useful ones.
9. **How is channel attention different from spatial attention?**
   Channel attention weights feature channels, answering "what features are important?" Spatial attention weights locations, answering "where is the useful information?"
10. **Why can channel and spatial attention be used sequentially?**
    Channel attention can first select useful feature types, and spatial attention can then highlight useful locations. Together, they refine both what the model uses and where it looks.
11. **Why might channel attention behave differently at `conv3_x` and `conv5_x`?**
    The stages contain different feature resolutions and meanings. `conv3_x` has detailed intermediate features, while `conv5_x` has more abstract, class-related features.
12. **Why can spatial attention be useful at an intermediate ResNet stage?**
    Intermediate stages contain both spatial detail and meaningful object parts. Spatial attention can highlight useful regions before later layers combine them into high-level features.
13. **Why is self-attention computationally expensive when the feature map has high spatial resolution?**
    Self-attention compares each spatial location with other locations. Many locations create many pairwise comparisons, increasing memory use and computation.
14. **What happens to the size of the self-attention matrix when the spatial resolution increases?**
    For a feature map with $H \times W$ locations, the matrix is $HW \times HW$. Increasing the number of locations therefore increases the matrix size quadratically.
15. **How does attention performance vary when the insertion position is changed?**
    Performance can improve or decrease depending on where attention is added. In the supplied results, P1 has the highest accuracy and P4 the lowest among the recorded P1–P4 runs; confirm the experiments were controlled before drawing a firm conclusion.
16. **How does attention at P2 differ visually from attention at P4?**
    P2 may respond to detailed edges, colours, and textures because its feature maps have higher resolution. P4 may highlight broader, more semantic regions; the actual maps must be inspected to confirm this.
17. **Does applying attention at multiple stages always improve classification performance?**
    No. Extra attention adds computation and may not improve the learned features. In the supplied results, P3+P4 and P2+P3+P4 have lower accuracy than the single P3 run.
18. **Does Channel→Spatial attention behave differently from Spatial→Channel attention?**
    Yes, the order changes which features are refined first. The recorded Channel→Spatial P3 run has higher accuracy and macro F1 than Spatial→Channel, assuming other settings were the same.
19. **How does the spatial-attention kernel size influence the attention map?**
    Kernel size controls the neighbourhood used to calculate spatial weights. A larger kernel can include broader context, but does not guarantee better accuracy; the supplied results include only 3×3 and 5×5, not 7×7.
20. **How does SE attention differ from ECA attention?**
    SE uses fully connected layers with a bottleneck to learn channel weights. ECA uses a lightweight one-dimensional convolution across nearby channels without the same dimensionality-reduction bottleneck.
21. **How does CBAM combine channel and spatial attention?**
    CBAM applies channel attention followed by spatial attention. It first weights important channels and then weights important spatial locations.
22. **How does self-attention differ from channel attention?**

Channel attention learns the importance of feature channels. Self-attention models relationships between spatial locations so each location can use information from other locations.

23. **Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?**
    No. More parameters increase complexity but do not guarantee better predictions. Accuracy must be compared with parameter count and inference cost using measured results.
24. **What is the difference between an internal attention map and Grad-CAM?**
    An internal attention map is produced by an attention module during the forward pass. Grad-CAM uses gradients for a chosen class to highlight regions that contribute to a prediction.
25. **Why is Grad-CAM called a post-hoc explanation technique?**
    Grad-CAM is generated after a trained model makes a prediction. It uses gradients to explain the prediction rather than directly guiding the forward pass.
26. **Does the attention model suppress irrelevant background regions?**
    The metric table alone cannot confirm this. Inspect the spatial-attention overlays and Grad-CAM figures to see whether background activation is reduced compared with the baseline.
27. **Are the attention maps of correctly classified and incorrectly classified images different?**
    They may differ: incorrect predictions may focus on misleading background regions or miss important flower parts. This must be checked by comparing visualizations for correct and incorrect test predictions.
28. **Which flower classes benefit most from attention?**
    The A3/P3 report has the highest class F1 for daisy (0.978947) and the lowest for roses (0.938144). To determine which classes benefit most, the baseline class-wise results are also needed.
29. **What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?**
    A confusion matrix shows which classes are predicted correctly and which classes are confused with one another. Accuracy gives only the overall proportion of correct predictions.
30. **Why is macro F1 useful in addition to accuracy?**
    Macro F1 calculates F1 for each class and averages the results equally. It can reveal poor performance on individual classes that overall accuracy may hide.
31. **What is meant by the accuracy–complexity trade-off?**
    It is the balance between accuracy and model cost, such as parameters, memory, and computation. A small accuracy gain may not justify a much more expensive model.
32. **What is meant by the F1–inference-time trade-off?**
    It compares class-balanced predictive performance with the time needed to make predictions. A model with higher F1 may be less practical if it is much slower.
33. **Why must every ablation study use the same dataset split?**
    Using the same samples makes the models' evaluation comparable. Otherwise, differences may come from easier or harder data rather than the change being tested.
34. **Why must the same training protocol be maintained when comparing different attention mechanisms?**
    Consistent preprocessing, data splits, epochs, optimizer, and other settings make the comparison fair. Ideally, only the attention mechanism under study should change.
35. **What conclusions can be drawn from the Grad-CAM comparison?**
    Grad-CAM can show whether the baseline and attention model focus on different regions for the same prediction. Claims about better flower localization or background suppression must be based on the actual maps.

38. Observations to be Recorded

Students should record the following during experimentation:

1. Input and output feature dimensions for each attention module.
2. Training accuracy.
3. Validation accuracy.
4. Testing accuracy.
5. Training loss.
6. Validation loss.
7. Macro precision.
8. Macro recall.
9. Macro F1-score.
10. Number of trainable parameters.
11. Total number of model parameters.
12. Average inference time.
13. Confusion matrix.

14. Channel-attention weights.
15. Spatial-attention maps.
16. Self-attention responses.
17. Grad-CAM maps.
18. Attention maps for different insertion positions.

39. Suggested Laboratory Report Structure

The laboratory report may be organized as follows:

1. **Aim**
2. **Dataset Description**
3. **ResNet50 Architecture**
4. **Attention Mechanisms**
5. **Proposed Experimental Architecture**
6. **Experimental Setup**
7. **Attention-Type Ablation**
8. **Attention-Position Ablation**
9. **Performance Results**
10. **Complexity Analysis**
11. **Training and Validation Graphs**
12. **Confusion Matrix Analysis**
13. **Grad-CAM Analysis**
14. **Internal Attention-Map Analysis**
15. **Qualitative Comparison**
16. **Additional Attention-Mechanism Comparison**
17. **Discussion**
18. **Conclusion**

40. Discussion Guidelines

Students should not merely reproduce numerical results.

The discussion should explain:

- why a particular attention mechanism produced a change in performance;
- how attention location influences feature refinement;
- how feature resolution changes attention behavior;
- how model complexity changes with different mechanisms;
- whether performance improvements justify additional computational cost;
- whether attention maps correspond to meaningful object regions;
- whether attention helps suppress the background;
- whether attention behaves differently for incorrect predictions; and
- whether quantitative and qualitative observations support each other.

Students should avoid making conclusions based on a single image.

Qualitative observations should preferably be based on several representative test images.

41. Expected Outcome

At the end of the experiment, students should understand that an attention mechanism is not an isolated module whose behavior is independent of the backbone.

Its effectiveness depends on:

$$\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity}.\end{aligned}$$

Students should also understand that classification performance alone does not provide a complete explanation of an attention mechanism.

A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions must be based on the student's experimental results rather than on assumptions about which attention mechanism should perform best.

Source Code

GitHub Repository:
https://github.com/Teena-Pranava/Deep-Learning-Lab

42. References

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